

Instructor Run Sheet

Using LLMs to Programmatically Extract and Curate Research Data

Instructor copy

Setup — display on slide

1. Create folder: `data curation workshop resources 30 June 2026`
 2. Download `plants_site_A.csv`, `plants_site_B.csv`, `plants_sites_AB_large.csv` from Zoom chat
 3. Open Copilot at copilot.microsoft.com
 4. Open Word — title: `Prompt log – data curation workshop – 30 June 2026`
 5. Open data in Excel
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Segment 1 — Intro + framing

1.1 Welcome and logistics

1.2 The data curation problem

1.3 Scenario walkthrough

Segment 2 — Block 1: Data cleaning

2.0 Stage 1 — Data exploration

Drag and drop `plants_site_A.csv` into Copilot.

Prompt →

I am a researcher studying the effect of weed management on native plant restoration across two sites in West Auckland. I want to understand whether weeding alongside native planting (Site A) produces better plant outcomes than native planting alone (Site B).

Attached is my data set (`plants_site_A`)

Tell me what you understand about this data – the columns, and what they seem to measure.

Stage 2 — Numeric: `Height_cm`

Steps 1-2: Identify

Prompt →

Look at ``Height_cm`` only. Return a table with the mean, min, max, range.

Prompt →

Look at ``Height_cm`` only. Return a two-column table with the original column (``Hieght_cm``), and a new column (``Height_cm_cleaned``) with potential outlier values marked with "check".

Step 3: Fix

Prompt →

Modify value 340 to 34 and return the two-column table, do not include any notes in the new cell, jsut "34".

Step 4: Validate — paste into `plants_site_A`; check first, last, changed rows

Step 5: Log — copy prompt to Word table | column | prompt | summary of changes |

Step 6: Apply

Prompt →

Provide me the same summary stats and outlier checks for ``Num_Leaves`` and ``Num_Flowers``.

Prompt →

Modify values and return four column table (original column + `_cleaned``):
``Num_Leaves``: 847 -> 87.
``Num_Flowers``: -3 -> 3. Keep 22 as 22.

2.3 Categorical: `Site`, `Location`, `Species`

Steps 1-2: Identify — Excel: Data → Filter, check `Species` dropdown

Step 3: Fix

Prompt →

Clean categorical columns ``Site``, ``Location``, ``Quadrat``, ``Species`` and ``Flowering``.
I want them to be consistently spelt, all values lower-case, and values separated by underscores "_".

Return pairs of columns for each variable: the original and a new column ``[column_name]_cleaned`` for

each. In the cleaned column have the new, corrected value and use "check" if you are unsure.

Step 3b (optional): Condition

Prompt →

For `Condition`, return the original column and a new column `Condition\_cleaned` with suggested categories representing a consistent five-point scale from poor to excellent.

Step 5: Log — copy prompts to Word

2.4 Block 1 close

Segment 3 — Block 2: Feature extraction

3.1 Plant_Age_Months — guided

Prompt →

Look at `Date\_Planted` and `Date\_Measured`. These columns contain dates in inconsistent formats — some DD/MM/YYYY, some D/M/YY, some DD-Mon with no year, and possibly some US-format (M/D/YY).

Without standardising the formats first, calculate the age of each plant in whole months at the time of measurement. Add this as a new column called `Plant\_Age\_Months`. Return these three columns as a table. Flag any row where you cannot confidently interpret the dates with "check".

Validate — paste into spreadsheet; check row count, spot-check 3 rows, flag negatives

Step 5: Log

3.2 Plant_ID — independent

Prompt → *(hint if needed)*

Using `Site\_cleaned`, `Location\_cleaned`, `Quadrat\_cleaned`, and `Species\_cleaned`, create a new column `Plant\_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location = 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.

Validate — sort by `Plant_ID` in Excel; check all values unique

Step 5: Log

3.3 Notes — independent

Prompt →

I am about to analyse `Plant\_Age\_Months`, `Height\_cm`, and `Condition` for these plants. For each row, I want to know whether the `Notes` field gives any reason this row's measurements might be unreliable or need special handling.

Return two new columns: `Data\_Quality\_Flag`: Y or N; and `Data\_Quality\_Notes`: a short reason for any Y.

Prompt →

Provide me a row-by-row thematic analysis of the `Notes` field, limit your analysis of each comment into one of three, short distinct themes from the perspective of a botanist that you will decide. Include the analysis as column `Notes\_themes`.

Step 5: Log

3.5 Apply to Site B

Fresh Copilot conversation. Participants work from their Word log.

Save combined output as `plants_combined.csv`

Segment 4 — Wrap-up

4.1 Scale test

New Copilot conversation. Drag and drop `plants_sites_AB_large.csv`.

Prompt →

Here is a dataset of 1,520 plant survey records from two sites.
> What I want you to do is to complete all of the same steps you completed for site_A and _B. Return a table of all original columns (even those not modified) and `[column\_name]\_cleaned` for any new, cleaned ones.

> For numerical values, flag any potential outliers with "check"; for categorical variables, change spelling, separation errors, and flag any values you are uncertain about with "check".

> Perform these steps:

> Cleaned or new columns Modification prompt
Height_cm Look at `Height\_cm` only. Return a two-column table with the original column (`Height\_cm`), and a new column (`Height\_cm\_cleaned`) with potential outlier values marked with "check".
Num_Leaves
Num_Flowers Provide me the same summary stats and outlier checks for

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`Num_Leaves` and `Num_Flowers`.

> Notes_themes Provide me a row-by-row thematic analysis of the `Notes` field,
limit your analysis of each comment into one of three, short distinct themes from
the perspective of a botanist that you will decide. Include the analysis as
column `Notes_themes`.

> Site`Location`, `Quadrat`, `Species` and `Flowering`. Clean categorical
columns `Site`, `Location`, `Quadrat`, `Species` and `Flowering`.
I want them to be consistently spelt, all values lower-case, and values
separated by underscores "_".
Return pairs of columns for each variable: the original and a new column
`[column_name]_cleaned` for
each. In the cleaned column have the new, corrected value and use "check" if
you are unsure.

> Using `Site_cleaned`, `Location_cleaned`, `Quadrat_cleaned`, and
`Species_cleaned`,
create a new column `Plant_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location
= 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.

> Plant_Age_Months Look at `Date_Planted` and `Date_Measured`. These columns
contain dates
in inconsistent formats – some DD/MM/YYYY, some D/M/YY, some DD-Mon
with no year, and possibly some US-format (M/D/YY).
Without standardising the formats first, calculate the age of each plant
in whole months at the time of measurement. Add this as a new column
called `Plant_Age_Months`. Return the these three columns as a table.
Flag any row where you cannot confidently interpret the dates with "check".

> Condition For `Condition`, return the original column and a new column
`Condition_cleaned` with suggested categories representing a consistent five-
point scale from poor to excellent.

> Data_Quality_Flag
Data_Quality_Notes I am about to analyse 'Age_Months`, `Height_cm`, and
`Condition` for these
plants. For each row, I want to know whether the `Notes` field give any reason
this row's
measurements might be unreliable or need special handling.
Add two new columns: `Data_Quality_Flag`: Y or N; and `Data_Quality_Notes`: a
short reason for any Y."

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4.2 AI R Code

Prompt →

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Produce the R code to run these steps myself in one block of code (one script) I
can copy and paste across in one go. Put some thought into this.

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4.3 Decision Matrix

4.4 Next Steps

4.4 Considerations

4.5 Resources + close